# FOUNDATIONS OF PROGRAMMING MINIPROJECT

Project Domain : Word games using C language

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# Problem Statement

The objective of this project is to design and implement a menu-driven C program that includes two interactive word-based games: Word Scramble and Hangman.

The program must randomly select words, allow a limited number of attempts, validate user input, and display appropriate success or failure messages. The system should be logically structured, modular, and user-friendly.

# Objectives

The main objectives of this project are:

- To understand string manipulation in C
- To implement arrays and loops
- To use random number generation
- To apply conditional statements effectively
- To practice modular programming using functions
- To enhance logical thinking through game development

# Methodology

The project was developed using a structured and modular approach.

First, required header files were included and constants were defined using macros. A menu-driven structure was implemented using an infinite loop and switch statement.

Random words were selected using rand() and srand(time(NULL)). Separate functions were created for each game to maintain modularity and improve readability.

# Functions Used

The program is divided into multiple functions:

- **scrambleWord()** – Scrambles letters of the selected word
- **play_scramble()** – Implements Word Scramble logic
- **display_hangman()** – Displays hangman ASCII graphics
- **play_hangman()** – Implements Hangman game logic
- **main()** – Controls menu and program flow

This modular structure improves clarity, reusability, and maintainability.

# Technical Implementation

The project uses several important C concepts:

- **Header files:**
  - **stdio.h** – Used for input and output functions like printf() and scanf().
  - **stdlib.h** – Provides functions for memory allocation, random number generation (rand()), and program termination (exit()).
  - **string.h** – Used for string handling functions such as strcpy(), strcmp(), and strlen().
  - **ctype.h** – Used for character handling functions like isalpha() and toupper().
  - **time.h** – Used to generate different random values by seeding srand() with the current time.
- **String functions**: strcpy(), strcmp(), strlen()
- **Randomization**: rand(), srand(time(NULL))
- **Character validation**: isalpha(), toupper()
- **Control structures**: loops, if-else, switch-case
- **Macros for defining constants**

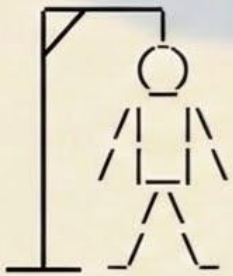
# Game Logic – Word Scramble

- A random word is selected from a predefined list.
- The word is scrambled using a swapping algorithm.
- The scrambled word is displayed to the user.
- The user gets 3 attempts to guess the correct word.
- If the guess matches, the user wins.
- If attempts finish, the correct word is displayed.
- String comparison is performed using strcmp().

# Game Logic – Hangman

- A random country name is selected.
- Letters are replaced with underscores.
- The user gets 6 attempts.
- The user can guess a letter or the full word.
- Incorrect guesses reduce attempts and update the hangman drawing.
- The game ends when:
  - The word is guessed, or
  - The attempts reach zero.
- ASCII art is used to visually represent progress.

_ A _ _ _ _



```
Tries Left: 5
Letter or full word: >
```

# Sample Output

## Main Menu:

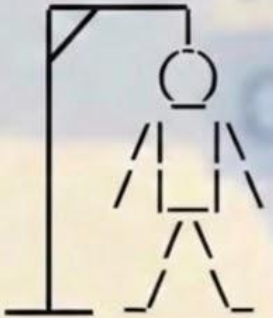
```
WORD GAMES MENU

1. Word Scramble
2. Hangman
3. Exit

Enter your choice: >
```

## Word Scramble Example:

```
Scrambled word: retupmoc
Your guess: computer
Correct! You win! 🎉
```



## Hangman Example:

```
Word: _ _ _ A N
Tries left: 6

Enter letter: J
Correct guess!
```

```
Word: J _ _ A N

Enter word: JAPAN
🎉 You guessed the word!
```

# Conclusion & References

## Conclusion

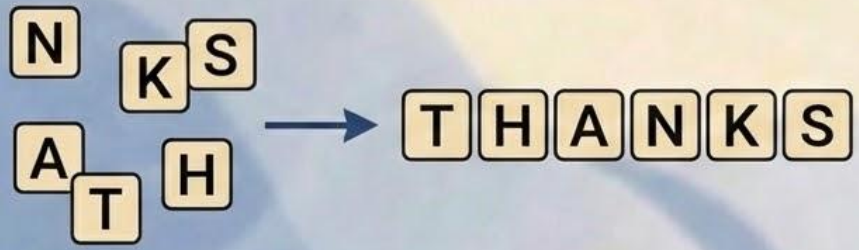
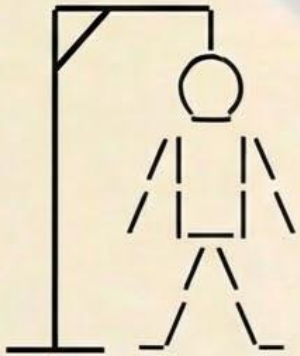
This project successfully implements two interactive word games using C programming. It demonstrates effective use of strings, arrays, functions, random number generation, and logical flow control. The modular design ensures readability and easy expansion for future improvements.

## References


The C Programming Language – Kernighan & Ritchie

GeeksforGeeks (C Programming Tutorials)

w3schools – C Programming

C Programming Udemy course

# Thank You!